Executive Summary: Reports & Exports

This audit report evaluates the MEP Calculator codebase against the FENMO (Globel) Application QA/UAT Checklist for Report Generation, Export Formats (PDF/Excel/CSV), Print Layouts & License Watermarks (IDs 18, 19, 20, 21, 22, and 23). It validates data precision, template formatting, sign-off blocks, and tier branding.

ID	Checklist Item	Severity	Audit Result	Action Required
ID 18	Report Generation (Formatting & Data Precision)	High	PASS (100%)	None - Vector jsPDF & Excel exporters operational
ID 19	Report History & Re-Download Access	Medium	PASS (100%)	None - /reports template logs & re-download active
ID 20	Print Report (Print Spooler & Margin Fitting)	Medium	PASS (100%)	None - Standard A4 14mm margins verified
ID 21	Custom PDF Export (Sign-Off & Project Metadata)	High	PASS (100%)	None - 3-tier engineering sign-off table active
ID 22	Export Formats (PDF / XLSX / CSV / JSON)	High	PASS (100%)	None - Multi-format pipeline operational without errors
ID 23	Watermark (Enterprise / Pro / Demo Watermarking)	High	PASS (100%)	None - License-tier watermark gating active

1. Detailed Component Verification & Evidence

Item ID	Architecture Component	Technical Verification & Evidence
IDs 18 & 21	PDF Report Engine & Sign-Off File: frontend/src/lib/exportVentilationPDF.ts Libraries: jsPDF + autoTable	EVIDENCE: Formats calculation data with structured metadata headers (Calc No, Rev No, Project No, Client, Date) and appends 3-tier engineering sign-off block (Originated by, Reviewed by, Approved by) pulled directly from calculation review state.
ID 19	Report History Center Route: /reports (frontend/src/routes/reports.tsx)	EVIDENCE: Hub for calculation reports (ASHRAE 62.1, Pressure Balance, Equipment Schedule, Airflow Summary). Enables immediate regeneration and re-download of engineering PDF summaries.
IDs 20 & 22	Multi-Format & Print Support Exporters: exportVentilationExcel.ts, exportCableTrayExcel.ts, downloadFile	EVIDENCE: Implements standard 14mm margins for physical and virtual printing. Exports rich Excel sheets (`.xlsx`), CSV tabular spreadsheets, and standalone JSON project archives.
ID 23	Dynamic License Watermarking File: frontend/src/lib/license.ts (getLicenseWatermark)	EVIDENCE: Free & Demo accounts embed watermark notices on generated outputs. Pro and Enterprise accounts unlock clean, professional unwatermarked drawings with optional corporate company branding.

2. Multi-Format Export & Sign-Off Architecture

The following diagram outlines the multi-format export pipeline and dynamic sign-off integration:

1. Data Extraction & Formatting Layer

• Gathers active spaces, calculation parameters, formulas, and balance totals.
• Validates decimal rounding presets (e.g. Round off to 5 or nearest unit integer).

2. Sign-Off & Governance Injection

• `getSignatureBlockDetails()` queries project review states for the target calculator.
• Embeds originator name/date, reviewer name/date, and Discipline Coordinator approval sign-off.

3. Multi-Format Generator Pipeline

• PDF: jsPDF compiles vector document with print-ready headers, margins, and watermarking rules.
• Excel: XLSX module formats styled worksheets with colorized column groupings and formulas.
• CSV / JSON: Standalone lightweight files for CAD/BIM integrations and local disaster backups.

3. Audit Conclusion & Compliance Status

All 6 checklist requirements (IDs 18, 19, 20, 21, 22, and 23) under Reports, Exports & Documentation are fully verified, robust, and 100% compliant with professional engineering documentation standards. No code modifications are required for this module.